

Weekly Update 4: [Nov. 05 - 12]

Developing a complete TinyML pipeline for indoor localisation using WiFi RSSI signals from residential wearable datasets. The pipeline includes data exploration, preprocessing, model training, and quantization for deployment on resource-constrained edge devices.

Completed Work

1. Dataset Processing

- **Source:** Residential wearable WiFi RSSI data across multiple houses and experiments
- **Format:** `rx_wearable_data.dat` (sensor readings) + `tag_annotations.dat` (location labels)
- **Processing Pipeline:**
 - Implemented timestamp-based alignment with 5-second tolerance using `merge_asof`
 - Added fallback index-based alignment for datasets without timestamps
 - Pivoted long-format data (AP addresses) to wide-format feature matrix
 - Handled missing values with -120 dBm imputation
- **Output:** Combined dataset with labeled samples from all experiments

2. Data Exploration & Analysis

Generated comprehensive visualizations and statistics:

- RSSI signal heatmaps showing spatial patterns across access points
- Class distribution analysis revealing location imbalance ratios
- Cross-house comparison of sample distributions
- Missing data analysis per access point
- **Deliverables:** 4 visualization plots, summary JSON, preprocessed pickle file

3. Model Training Pipeline

Preprocessing:

- Window-based feature extraction (4-second windows, 2-second stride)
- StandardScaler normalization after imputation
- Train/Val/Test split: 75%/10%/15% (stratified)

Architectures Implemented:

- **MLP:** 3-layer feedforward with dropout (128→64→classes)
- **LSTM:** 2-layer bidirectional LSTM (64 hidden units) with dropout

Training Configuration:

- Batch size: 64, Epochs: 25
- Adam optimizer (lr=1e-3) with ReduceLROnPlateau scheduler
- CrossEntropyLoss with weight decay (1e-5)

4. Model Quantization (TinyML Optimization)

- Applied PyTorch dynamic quantization (FP32 → INT8)
- **Results:** ~4x model size reduction with minimal accuracy loss (<2%)
- Quantized both MLP and LSTM models for edge deployment
- **Outputs:** FP32 and INT8 model checkpoints (.pth files)

Key Results

Model	Test F1(FP32)	Test F1(INT8)	Size Reduction
MLP	0.1016	0.1029	~75%
LSTM	0.0626	0.0652	~75%

Next Steps

1. **Knowledge Distillation:** Test teacher-student framework to further compress models while maintaining accuracy
2. **Advanced Architectures:** Explore Transformer and Mamba-based sequence models for improved performance
3. **Edge Deployment:** Port quantized models to Edge Impulse Studio and Arduino Nano 33 BLE
4. **Benchmarking:** Compare performance, latency, and memory footprint on actual edge hardware

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